Javier Díez Pérez

AI ENGINEER, INNOVATOR & AGENTIC SYSTEMS ARCHITECT

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CORE SKILLS

AI & Agents
Multi-agent orchestration (Agno, ADK, LangGraph), MCP servers, RAG, Multimodal RAG, GraphRAG, RIG, Tool-use architectures, Agent evaluation & observability, Specs-driven development, Human-in-the-loop design, LLM application security, Claude Code plugins

Data Science & ML
scikit-learn, XGBoost, PySpark, Dask, Ray, H2O, SHAP, lime, Pandas, Polars

Knowledge Graphs
RDF/LPG, SPARQL, Neo4j, Ontologies, Causal KGs, Biomedical KGs, GraphRAG, Data Modelling

Programming
Python, R, SQL, Scala, Rust, HTML/CSS

Data Stores
PostgreSQL, MySQL, DuckDB, MongoDB, Hive, Elasticsearch, Vector DBs

Pipelines & CI/CD
Prefect, Airflow, Nextflow, Snakemake, GitHub Actions, pytest, Docker

Cloud & DevOps
AWS (Big Data & ML), GCP, Terraform, Packer

Bioinformatics
Bioconductor, Biopython, NGS pipelines, GATK, SAMTools, scanpy

Statistics & GIS
Experiment design, multivariate analysis, PostGIS, QGIS

LANGUAGES

Spanish	Native
English	Proficient
French	Good
Portuguese	Basic

PROFILE

Driven by the intersection of science and technology, I operate as a bold innovator who turns complex research into disruptive, scalable realities. I define the product vision, strategy, and roadmap for next-generation, multi-agentic platforms that challenge traditional industry workflows. Operating at the convergence of advanced machine learning, empirical research, and business development, I translate complex multimodal AI capabilities into high-value commercial tools. I am adept at navigating ambiguity, validating user needs in uncharted markets, and managing ethical guardrails around sensitive data. By driving orchestration-focused product design, I ensure we deliver compliant, high-impact products that solve genuine bottlenecks.

EXPERIENCE

Independent consultant - Barcelona
2026 - Present

AI engineer building agentic innovation systems across Biotech amd Healthcare. As an AI-native innovator, I architect innovative, AI-native products from the ground up using Python, Agno, and Google ADK.
While my technology is deeply technical, my philosophy is profoundly people-centric and patient centric.
By anchoring LLMs to Knowledge Graphs and Causal Graphs, I ensure that my autonomous agents deliver deterministic, safe, and transparent outcomes that directly accelerate R&D pipelines, optimize clinical trials, and ultimately improve patient care.

AstraZeneca · Barcelona
2023 — 2026

Senior Innovation Specialist DS & AI (ODSAI, 2025–Present)

- Built agentic conversational app for structured/unstructured data (RAG/RIG) with multi-agent architecture using Agno and Google ADK frameworks
- Developed competitive intelligence multi-agent application following Strategyzer methodology
- Built MCP servers for enterprise interaction with structured and unstructured data sources
- Developed multimodal RAG for enterprise knowledge retrieval
- Alexion 2025 spring accelerator — innovation modelling via Strategyzer's framework

Data Product Lead — Oncology DS Platforms (2023–2025)

- Led data product operations for PDx mouse models and ENABL platform
- Designed and maintained oncology data science platform infrastructure

Data Engineer & Python Developer
2022 — 2023

Contractor · Madrid

- Data products for pharma (GSK) and food (HdR); Python packages for data validation & FAIRification
- GCP ETL pipeline development with integrated validation framework

Lead Bioinformatician & Data Scientist
2021 — 2022

PharmaMar · Madrid

- R&D bioinformatics lead: cancer genomics, drug development, biomarker discovery
- Multiomics (DNA-seq, RNA-seq, ChIP-seq, ATAC-seq) and clinical trial data integration

Senior Bioinformatician & Data Scientist
2020 — 2021

Fujitsu · Madrid

- Digital transition at European CoE for Data Intelligence — EHR APIs, SNOMED-CT, entity recognition

Senior Bioinformatics Scientist
2019 — 2020

HiFiBio · Paris

- Single cell RNA-seq analysis for TCR in TILs; sc-targeted RNAseq pipeline development
- Algorithm development for cell barcode UMI assignment and sequence correction
- Unsupervised ML data analysis for cell-type and TCR chain-type characterization

Data Analyst & Python Developer
2017 — 2020

Freelancer · Madrid

- Predictive analytics tools for banking, insurance, and telco industries
- Big data science and ML with Python and R ecosystem; package and tool development

Senior Bioinformatician
2016

Wellcome Trust Sanger Institute · Cambridge, UK

- Somatic mutation signatures on DNA repair defective cell lines
- Whole genome sequencing analysis of paired normal/tumor samples: SNV, Indel, CNVs
- Whole exome sequencing analysis from TCGA (The Cancer Genome Atlas)

Bioinformatics Analyst
2015

Congenica Ltd. · Cambridge, UK

- Splicing mutation effect analysis, mutation-created splice sites and cryptic splice sites
- Bioinformatic analysis of trio and family from NGS data

VOLUNTEERING

CRAM Marine Animals Rehab (2024–25)

Madrid4Refugees coding teacher

Save the Children tutor

Bioinformatician — Molecular Pathology

2014 — 2015

Royal Marsden Hospital NHS Foundation Trust · London, UK

- NGS pipelines for tumor molecular pathology: panel targeted, immunoglobulin/TCR, amplicon analysis
- Amplicon molecular barcoding deduplication pipeline
- Study Bioinformatician at FoRMAT (molecular characterisation of advanced GI tumors)

Bioinformatician — Clinical Neurogenetics

2013 — 2014

UCL / National Hospital for Neurology · London, UK

- NGS pipelines for clinical neurogenetics: exome, panel targeted, mtDNA heteroplasmy analysis
- Database development for clinical information, diagnostic test results, and annotation

Field Bioinformatics Specialist

2012 — 2013

Life Technologies · Mexico DF, Mexico

- Set up bioinformatic support and training for Ion Torrent (PGM/Proton) and SOLiD platforms across Latin America
- Project design, troubleshooting for whole genome, resequencing, and targeted sequencing

Bioinformatician — Epigenetics & Cancer Biology

2011 — 2012

IDIBELL (Bellvitge Biomedical Research Institute) · Barcelona

- EPINORC project: Epigenetic Disruption of Non-Coding RNAs in Human Cancer (ERC Advanced Grant)
- Methylation arrays, bisulphite sequencing, WGS, exome NGS, RNA-seq, ChIP-seq analysis

Bioinformatics Research Assistant

2008 — 2010

CRG / Parc de Recerca Biomèdica de Barcelona

- De novo genome assembly, resequencing and genome analysis of Saccharomyces cerevisiae
- Developed DeathBase: relational database and web front-end for cell death curated proteins
- Characterization of putative human BH3-only proteins; cancer susceptibility gene analysis

Research Assistant

2005 — 2008

CSIC (CNB & CBM-SO) · Madrid

- Human centrosome proteome study and CentrosomeDB database development
- Cell death genetics in Drosophila; ML approach for phenotype trait analysis

EDUCATION

Specialist — Quantitative Research & Statistics

2006 — 2007

Polytechnic University of Madrid & CSIC

DEA / Master — Biochemistry & Molecular Biology

2005 — 2007

Universidad Autónoma de Madrid

B.Sc. Biochemistry

2001 — 2004

Universidad Autónoma de Madrid

SELECTED PUBLICATIONS

PEER-REVIEWED PUBLICATIONS

- Costanzo F, Martínez Diez M, Santamaría Nuñez G, Díaz-Hernández JI, Genes Robles CM, **Díez Pérez J**, Compe E, Ricci R, Li TK, Coin F, Martínez Leal JF, Garrido-Martín EM, Egly JM. "Promoters of ASCL1- and NEUROD1-dependent genes are specific targets of lurbinectedin in SCLC cells." *EMBO Mol Med*, 2022. PMID:35263037
- Jiménez-Guardeño JM, Ortega-Prieto AM, Menéndez Moreno B, Maguire TJA, Richardson A, Díaz-Hernández JI, **Diez Perez J**, Zuckerman M, et al. "Drug repurposing based on a quantum-inspired method versus classical fingerprinting uncovers potential antivirals against SARS-CoV-2." *PLoS Comput Biol*, 2022. PMID:35849631
- Glodzik D, Morganella S, Davies H, ..., **Diez-Perez J**, ..., Nik-Zainal S. "A somatic-mutational process recurrently duplicates germline susceptibility loci and tissue-specific super-enhancers in breast cancers." *Nature Genetics*, 2017. PMID:28112740
- Galindo-Moreno J, Iurlaro R, El Mjiyad N, **Díez-Pérez J**, Gabaldón T, Muñoz-Pinedo C. "Apolipoprotein L2 contains a BH3-like domain but it does not behave as a BH3-only protein." *Cell Death Dis*, 2014. PMID:24901046
- Heyn H, Li N, Ferreira HJ, Moran S, Pisano DG, Gomez A, **Diez J**, ..., Esteller M. "Distinct DNA methylomes of newborns and centenarians." *PNAS*, 2012. PMID:22689993
- Neurohr G, Naegeli A, Titos I, Theler D, Greber B, **Díez J**, Gabaldón T, Mendoza M, Barral Y. "A midzone-based ruler adjusts chromosome compaction to anaphase spindle length." *Science*, 2011. PMID:21393511
- **Díez J**, Walter D, Muñoz-Pinedo C, Gabaldón T. "DeathBase: a database on structure, evolution and function of proteins involved in apoptosis and other forms of cell death." *Cell Death Differ*, 2010. PMID:20383157
- Nogales-Cadenas R, Abascal F, **Díez-Pérez J**, Carazo JM, Pascual-Montano A. "CentrosomeDB: a human centrosomal proteins database." *Nucleic Acids Res*, 2009. PMID:18971254
- Bonifaci N, Berenguer A, **Díez J**, Reina O, Medina I, Dopazo J, Moreno V, Pujana MA. "Biological processes, properties and molecular wiring diagrams of candidate low-penetrance breast cancer susceptibility genes." *BMC Med Genomics*, 2008. PMID:19094230

PRESENTATIONS & CONFERENCES

- Parmar C, Roudko V, Muckerson V, Blando J, Daniel J, **Díez J**, Gunda S, Mendoza Alcala A, Franco A, Stambolic I, Benchea C, Stewart R, Lai Z, Reis-Filho J, Scaltriti M, Palmer D. "AI-Derived H&E Features Stratify Overall Survival and Suggest Differential Benefits in NEPTUNE (1L mNSCLC)." *AACR Annual Meeting*, San Diego, 2026
- Moorcraft SY, González de Castro D, Cunningham D, Walker BA, Hulkki Wilson S, **Díez Pérez J**, et al. "FOrMAT: Feasibility of a molecular characterization approach to treatment of patients with advanced GI tumors." *J Clin Oncol* 33, 2015 (ASCO). DOI
- McCall S, Polke JM, **Díez-Pérez J**, Gardiner A, Jaffer F, Nanji T, Pittman A, Hughes D, Hanna MG, Houlden H. "Next Generation Sequencing of Neuronal Channelopathies." *British Society for Genetic Medicine Annual Meeting*, Liverpool, 2013
- "ION PROTON DATA ANALYSIS" — Mexico & Latin America Ion Proton launch events, Life Technologies, 2012–2013
- ASHG Annual Meeting 2012, San Francisco · RECOMB::DREAM 2011, Bellvitge · Mono-allelic Expression in Health and Disease, Barcelona · Systems Biology & New Sequencing Technologies (SBNST), CRG Barcelona

Last updated: August 23, 2026